Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
: 101604

Subject: Engineering Economics, Estimation & Costing

Time: 03 Hours
Full Marks: 70

tions:-

The marks are indicated in the right-hand margin.

*There are **NINE** questions in this paper.*

*ii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): [2 x 7 = 14]

(a) If the price of a good increases and the quantity demanded decreases, this represents the law of;
(i) Supply
(ii) Diminishing returns
(iii) Demand
(iv) Elasticity

(b) Which of the following is a direct tax?
(i) Sales Tax
(ii) Income Tax
(iii) Value Added Tax (VAT)
(iv) Excise Duty

(c) Which institution primarily controls monetary policy in a country?
(i) Commercial Banks
(ii) The Treasury Department
(iii) The Central Bank
(iv) The Stock Exchange

(d) Which of the following was the primary goal of India's first Five-Year Plan (1951-1956)?
(i) Industrialization
(ii) Infrastructure development
(iii) Agricultural development
(iv) Poverty alleviation

(e) Which of the following is NOT a form of business organization?
(i) Sole proprietorship
(ii) Monarchy
(iii) Partnership
(iv) Corporation

(f) Net Present Value (NPV) refers to
(i) The value of future cash flows discounted to the present
(ii) The rate of return on investment
(iii) The future value of current investments
(iv) The interest rate for a loan

(g) In construction, Bar Bending Schedule is used for:
(i) Estimating concrete quantity
(ii) Labor cost estimation
(iii) Preparing a schedule for project phases
(iv) Determining the amount of steel reinforcement

(h) Why are specifications important in construction?
(i) They ensure quality and compliance with standards
(ii) They define contract payment terms
(iii) They provide a framework for estimation
(iv) They calculate project timelines

(i) Which factor does NOT affect rate analysis?
(i) Material cost
(ii) Labor cost
(iii) Weather conditions
(iv) Equipment costs

(j) Which type of contract allows for adjustments due to inflation?
(i) Lump sum contract
(ii) Cost-plus contract
(iii) Fixed price contract
(iv) Unit rate contract

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Q.2 (a) What is 'Disposable Income'? Explain in detail. [7]
(b) Define and differentiate Open and Closed Economies with examples. [7]

Q.3 (a) What are the causes of Child Labor? [7]
(b) What are the types of Monetary systems? [7]

Q.4 (a) Explain Cost and Cost control techniques in detail. [7]
(b) What do you mean by budgets? Explain application of break- even analysis. [7]

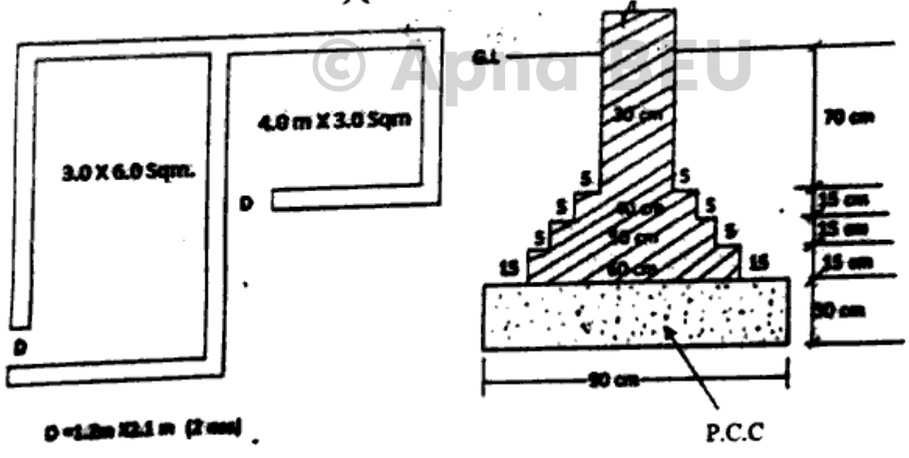
Q.5 (a) What are the objectives of third year plan? [7]
(b) Write a note on Informal Employment. [7]

Q.6 (a) Explain what is finishes, interior, and MEP work in detail. [7]
(b) Calculate the volume of earthwork for a Proposed Road having formation width 7.5 m and side slope 1:1 using mid sectional area Method. Details of road are given below.

Chainage (m)	0	30	60	90	120	150	180	210	240	270	300
G.L. (m)	10.55	10.5	10.4	10.35	10.2	10	9.8	9.65	9.6	9.4	9.2

Formation level of Road = 10.00 m

Q.7 (a) For the given, compute the quantities of following items. [7]
(1) Earthwork in excavation
(2) Cement concrete in foundation
(3) 1st class Brick masonry work up to G.L in c.m. (1:6)
(4) Back filling in foundation trench



Note: 1. Drawings are not to the scale
2. All walls are 30 cm thick

(b) Explain what are the requirements and importance of specifications. [7]

Q.8 (a) What is meant by Rate analysis? Explain the purpose of Rate Anlysis. [7]
(b) How can a contract be terminated? [7]

Q.9 (a) Explain Item rate contract with its advantages and disadvantages in detail. [7]
(b) Define Arbitration. What are the types of Arbitration? [7]

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